GREG DICRISTOFARO

Software Engineer

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SUMMARY

Software Engineer with 8+ years of experience designing, testing, and deploying full-stack applications across a wide range of domains. Expertise spans Java (Spring Boot), Python, C++, TypeScript, C# (ASP.NET Core), modern frontend frameworks (React, Angular), cloud infrastructure (GCP, AWS), and CI/CD practices (Docker, Jenkins, GitHub Actions). Former **educator of 11 years** with a proven track record of collaborating and translating complex technical concepts for non-technical audiences.

EXPERIENCE

Software Engineer

Sleuth Kit Labs

February 2020 - April 2026 *Somerville, MA*

- Developed **Digital Forensics and Incident Response** software: Cyber Triage, Autopsy, and The Sleuth Kit in an Agile/Scrum environment
- Expanded and optimized **data pipeline processing** including complex SQL queries over large data sets using Python, Java, and C++
- Created **web crawling and scraping** scripts aggregating open source data to enrich the data pipeline with new features
- Designed and implemented cloud updatable heuristics served to on-premises deployments via **GCP Datastore, BigQuery and Cloud Functions**, then compiled into optimized predicate matching data structures with runtimes in milliseconds for millions of files
- **Modernized testing and CI/CD infrastructure** using Jenkins, Docker, Linux, Bash, GitHub Actions, CircleCI, Python, JUnit/Mockito, SonarQube, and Playwright
- **Standardized multi-platform deployment (Windows, Linux, macOS)** through Snap, Homebrew, and Advanced Installer packages
- Authored internal and **customer-facing documentation** for complex features
- Architected and built **RESTful API and Reactive web applications** using Spring Boot, Swagger, Angular, and React (Material UI), leveraging Server Sent Events (SSE) for real-time data streaming
- Built a **file hash lookup web service** using AWS DynamoDB and Lambda for third-party integration
- Integrated **AWS S3** and **Azure Blob Storage** as data sources for file upload and retrieval
- Architected **parallel data ingestion system** using Docker, JMS, ActiveMQ, and AWS SQS for coordination, improving scalability of data processing
- Developed third-party integrations in **Go**
- Leveraged **Node.js** across web application development, desktop tooling (Tauri), and build utilities

SKILLS

Languages: Java, TypeScript, JavaScript, C#, Python, Bash, SQL, C, C++, HTML/CSS, Go

Web/Backend: React, Angular, Spring Boot, ASP.NET Core, Server Sent Events (SSE), WebAssembly, Node.js, Flask

Cloud/DevOps: GCP (BigQuery, Cloud Functions, Datastore), AWS (Lambda, DynamoDB, S3, SQS), Azure Blob Storage, Docker, Jenkins, SonarQube, Playwright, Linux

Databases: PostgreSQL, MS SQL Server, SQLite, MongoDB

Desktop: Tauri, JavaFX, Swing

Development/VCS Tools: Git, GitHub, Claude Code, Copilot, Antigravity, Visual Studio Code, Visual Studio IDE, IntelliJ IDEA IDE

ACHIEVEMENTS

✓ Presenter

Presenter for OSDFCon 2021 (Autopsy Scoring: Finding the Relevant Data with Analysis Results)

✓ KBR Certificate of Appreciation

In recognition of outstanding performance & commitment to excellence (2019)

PROJECTS

constantqjs

Implemented the Constant-Q transform for **spectral analysis** using C++ compiled to **WebAssembly** (via Emscripten), integrating with **TypeScript/Angular** for real-time visualization (Chart.js, Web Audio API)
gdicristofaro.github.io/constantqjs

Software Engineer

KBR

July 2017 - February 2020 *Lexington Park, MD*

- Developed **Single Sign On (SSO) and Role-Based Access Control (RBAC)** web application
- Queried large datasets using **Microsoft SQL Server, Entity Framework**, T-SQL, and Linq to SQL
- Built tooling to **generate parameterized queries** tailored to the deployment environment, improving query performance from minutes to milliseconds and eliminating SQL injection vulnerabilities
- Created **data visualizations** with JavaScript, TypeScript, **D3.js**, and SVG to present complex data to customers
- Led creation and deployment of a **React and ASP.NET Core** web application with CMS and Authentication/Authorization capabilities, collaborating with stakeholders and a UX designer
- Modernized large J2EE web applications built with JSP and JSF, earning an **award of excellence** for customer satisfaction

Music Teacher

St. Mary's County Public Schools

August 2006 - June 2017 *Leonardtown, MD*

- Built software tools for classroom use, including a **constraint-based scheduling system (CSPSolver)** to automate school scheduling
- Demonstrated strong organizational, leadership, and logistics skills by coordinating school-wide schedules and leading public-facing events
- Developed differentiated and scaffolded resources and curriculum to support student success at multiple skill levels

EDUCATION

Master of Science in Software Development

Boston University

September 2017 - May 2019 *Boston, MA* *GPA 3.96 / 4.00*

Software Engineering Certificate

Harvard University

September 2012 - December 2014 *Cambridge, MA* *GPA 4.00 / 4.00*

Master of Music in Music Education

University of the Arts

June 2012 - December 2015 *Philadelphia, PA* *GPA 4.00 / 4.00*

Designed cross-platform music composition software to foster creativity for young students that processed PCM audio data, visualized music in a sequencer, and rendered standard music notation

sci-fi-crawl

Sci-fi customizable opening credit text using **React, TypeScript, Material UI, and CSS animations**
gdicristofaro.github.io/sci-fi-crawl

PSDtoSVG

Python web application for converting Photoshop files into SVGs for data visualization
gdicristofaro.github.io/PSDtoSVG

SeamCarve

Scala implementation of the Seam-Carving Algorithm for content-aware image resizing
github.com/gdicristofaro/SeamCarve

CSPSolver

Class scheduling program written in Scala based on the AC-3 algorithm, importing data from Excel documents via Apache POI
github.com/gdicristofaro/CSPSolver

CalendarMaker

Calendar making application developed with TypeScript, React, Moment.js, Material-UI, and PptxGenJS
github.com/gdicristofaro/CalendarMaker